

Internal Assessment

Modeling the reflections of a spherical mirror

Mathematics Analysis and Approaches: HL

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1 Introduction

When I was having my morning coffee, I noticed that my reflection seen on the concave side of the spoon was upside down and deformed. This observation fascinated me because we are mainly accustomed to regular flat mirrors that show a rather accurate copy of a reflected object. If we assume that the mirror and the spoon are made from the same material that emits the light perfectly (e.g., the light is not scattered), the only difference between those 2 objects is in the type of curvature. Since the spoon is far too complicated shape to model, I considered a mirror shaped like a hemisphere, which can be described by a simple equation. The exploration will therefore aim to mathematically describe the reflection of light rays from a concave part of the spherical mirror using vectors and provide a visualization to better understand the projection seen on the mirror.

Similar process is already used in 3D graphics. According to Brookshear, "ray tracing is essentially the process of following a ray of light backward to find its source" [1, Ch. 10.5]. With ray tracing, several light phenomena can be simulated, namely, reflection, refraction, and scattering of the light. This rendering technique is able to produce hyper realistic images and therefore is used in the movie CGI and lately, in video games. In the exploration, the path of a given light ray will be traced in similar fashion from the camera (or the observer) onto a spherical mirror. Any changes in the direction vector of the light ray will be denoted.

2 Law of Reflection

The only physical phenomenon used in this mathematical exploration is the law of reflection, which states that the angle of incidence (angle between the light ray and normal to the surface) and angle of reflection (angle between the reflected light ray and normal to the surface) are equal [2, p. 316] and the light ray, normal and reflected ray lie on the same plane, as shown in figure 1.

In this section, relationship between the vectors of light ray, normal to the surface and reflected ray is to be derived. Suppose the original ray has a direction vector $\vec{u}$, the normal vector is $\vec{n}$ and the reflected ray has a direction vector $\vec{p}$, as shown in figure 2. B is the place of reflection. It is assumed that vectors $\vec{u}$, $\vec{n}$ and $\vec{p}$ are unit vectors. The law of

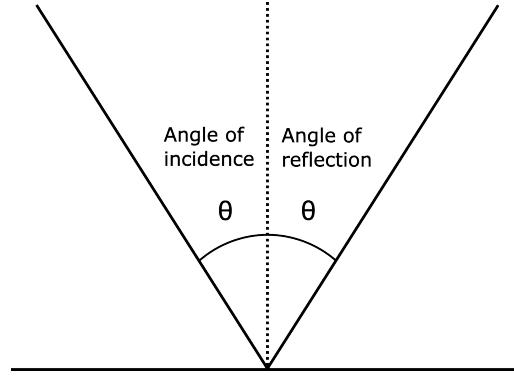


Figure 1: Law of reflection

reflection holds true, therefore, $|\angle \vec{u} \vec{n}| = |\angle \vec{n} \vec{p}| = \theta$. Let A and C be points such that $\overrightarrow{AB} \equiv \vec{u}$ and $\overrightarrow{BC} \equiv \vec{p}$ respectively. D is defined as an intersection of line AC and the normal line. It is clear that triangles $\triangle ABD$ and $\triangle CBD$ are congruent, since $|AB| = |CB|$, $|\angle ABD| = |\angle CBD|$, and both triangles share a segment DB . Therefore, segments AD and DC have the same length denoted by a vector $\vec{d}$.

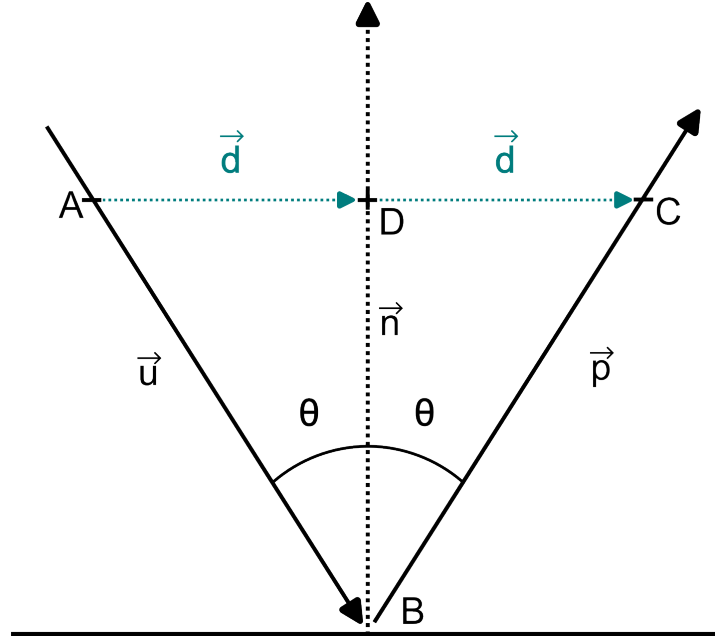


Figure 2: Law of reflection shown through vectors

Figure 2 illustrates, that vector $\vec{p}$ can be expressed as composition of vectors $\overrightarrow{BD}$ and $\vec{d}$. Vector $\overrightarrow{BD}$ is a scalar multiple of $\vec{n}$, since vectors $\overrightarrow{BD}$ and $\vec{n}$ have the same direction. $|\overrightarrow{BD}|$ can be expressed as $|AB|\cos\theta = \cos\theta$, since $|AB| = 1$. From properties of dot product we know that:

$$|\overrightarrow{BD}| = \cos\theta = \frac{-\overrightarrow{u} \cdot \overrightarrow{n}}{|\overrightarrow{u}| |\overrightarrow{n}|}$$

Recalling that $|\overrightarrow{u}| = |\overrightarrow{n}| = 1$:

$$|\overrightarrow{BD}| = -\overrightarrow{u} \cdot \overrightarrow{n}$$

Since $\overrightarrow{n}$ is a unit vector, we can conclude that $\overrightarrow{BD} = (-\overrightarrow{u} \cdot \overrightarrow{n}) \overrightarrow{n} = -(\overrightarrow{u} \cdot \overrightarrow{n}) \overrightarrow{n}$. As for the vector $\overrightarrow{d}$, it can be expressed as $\overrightarrow{u} + \overrightarrow{BD} = \overrightarrow{u} - (\overrightarrow{u} \cdot \overrightarrow{n}) \overrightarrow{n}$. Finally, we have all the information needed to express $\overrightarrow{p}$ in terms of $\overrightarrow{u}$ and $\overrightarrow{n}$:

$$\overrightarrow{p} = \overrightarrow{BD} + \overrightarrow{d}$$

$$\overrightarrow{p} = -(\overrightarrow{u} \cdot \overrightarrow{n}) \overrightarrow{n} + \overrightarrow{u} - (\overrightarrow{u} \cdot \overrightarrow{n}) \overrightarrow{n}$$

$$\overrightarrow{p} = \overrightarrow{u} - 2(\overrightarrow{u} \cdot \overrightarrow{n}) \overrightarrow{n}$$

which is relationship used throughout the rest of the exploration.

3 Describing Reflection of Light Rays

In this section, the way to describe a reflection of light ray from a surface is given. Furthermore, we will introduce notation for capturing this phenomena. Suppose, the source of light is an arbitrary point in 3D space, from which the light ray travels in certain direction vector. Eventually, the light ray reaches some point of the reflective surface, and its direction vector changes.

It is quite useful to encapsulate this phenomena into one function, as it allows for abstraction used in later stages of the exploration. We introduce a notation used for describing the reflection of a light ray from a surface illustrated in Figure 3 expressed in terms of source of the light $B = (B_1, B_2, B_3)$, direction vector of a ray coming from source $\overrightarrow{u} = (u_1, u_2, u_3)$, the point of reflection $S = (S_1, S_2, S_3)$ and direction vector of the reflected ray $\overrightarrow{p} = (p_1, p_2, p_3)$:

$$(S, \overrightarrow{p}) = \text{鏡}(B, \overrightarrow{u})$$

or the expanded form:

$$(S_1, S_2, S_3, p_1, p_2, p_3) = \text{鏡}(B_1, B_2, B_3, u_1, u_2, u_3)$$

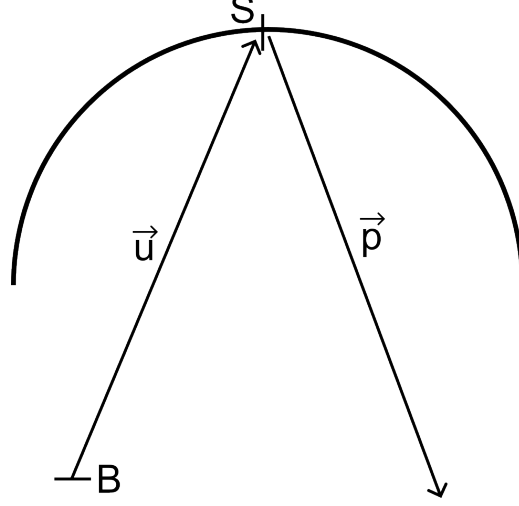


Figure 3: Visualization of $(S, \vec{p}) = \text{鏡}(B, \vec{u})$ notation on a 2D plane

鏡 was chosen for the function symbol, since it is a Japanese character for a mirror. To keep calculations simpler, $\vec{u}$ will always be a unit vector and it can be represented by direction cosines. Therefore $(u_1, u_2, u_3) = (\cos\alpha, \cos\beta, \cos\gamma)$, where α, β, γ are the angle between $\vec{u}$ and X, Y , and Z axis respectively.

4 Calculating the Point of Reflection on a Sphere

Before calculating an explicit formula for $\text{鏡}(B, \vec{u})$, we need a last prerequisite, an equation of a sphere. As shown in [3, p. 265], the equation for a sphere with center at a point $C = (a, b, c)$ and radius r is:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$$

In the exploration, we will consider a sphere with radius $r = 1$ and center $C = (0, 0, 0)$, with aim to simplify calculations. Furthermore, the mirror needs to be in a shape of hemisphere, so a restriction for x coordinates is added. This leaves us with the final equation for a spherical mirror h :

$$h : x^2 + y^2 + z^2 = 1, x > 0$$

The light ray coming from the source traces a line $\vec{l} = B + \mu \vec{u}, \mu \in \mathbb{R}$. To calculate an explicit formula for $\text{鏡}(B, \vec{u})$, we need to find the point of reflection on a hemisphere h , which is equivalent to the intersection of $\vec{l}$ and h :

$$\vec{l} = B + \mu \vec{u}$$

$$\vec{l} = \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix} + \mu \begin{pmatrix} \cos\alpha \\ \cos\beta \\ \cos\gamma \end{pmatrix}$$

this is equivalent to:

$$\vec{l} : \begin{cases} x = B_1 + \mu \cos\alpha \\ y = B_2 + \mu \cos\beta \\ z = B_3 + \mu \cos\gamma \end{cases}$$

After entering the values for x , y and z into h , we are able to calculate the scalar μ , using quadratic formula:

$$h : x^2 + y^2 + z^2 = 1$$

$$(B_1 + \mu \cos\alpha)^2 + (B_2 + \mu \cos\beta)^2 + (B_3 + \mu \cos\gamma)^2 = 1$$

$$B_1^2 + 2B_1\mu \cos\alpha + \mu^2 \cos^2\alpha + B_2^2 + 2B_2\mu \cos\beta + \mu^2 \cos^2\beta + B_3^2 + 2B_3\mu \cos\gamma + \mu^2 \cos^2\gamma = 1$$

$$(\cos^2\alpha + \cos^2\beta + \cos^2\gamma)\mu^2 + 2(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma)\mu + B_1^2 + B_2^2 + B_3^2 - 1 = 0$$

Since $|\vec{u}| = 1$, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$:

$$\mu^2 + 2(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma)\mu + B_1^2 + B_2^2 + B_3^2 - 1 = 0$$

After entering coefficients into quadratic formula and simplifying, we get the solution for μ :

$$\mu = \begin{cases} -(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma) - \sqrt{(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma)^2 - B_1^2 - B_2^2 - B_3^2 + 1} \\ -(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma) + \sqrt{(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma)^2 - B_1^2 - B_2^2 - B_3^2 + 1} \end{cases}$$

We will refer to those 2 solutions as μ_1 and μ_2 respectively. The only difference between μ_1 and μ_2 is that μ_1 subtracts the expression with square root from $-(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma)$, whereas μ_2 adds it. Since

$$\sqrt{(B_1\cos\alpha + B_2\cos\beta + B_3\cos\gamma)^2 - B_1^2 - B_2^2 - B_3^2 + 1} \geq 0$$

We can deduce that $\mu_1 \leq \mu_2$.

Although, there are 2 solutions for μ , we can intuitively deduce which one will be excluded. There are 3 cases; source of light is outside the sphere, inside or on its surface.

Assume, the source of light is outside as shown in figure 4. To be reflected from the concave surface of the sphere, which is inside the sphere, the light needs to come to contact with some other point of the sphere before. Since $\mu_1 \leq \mu_2$, it is clear the first point of intersection has coordinates $B + \mu_1 \vec{u}$ and therefore the point of reflection S must have coordinates $B + \mu_2 \vec{u}$.

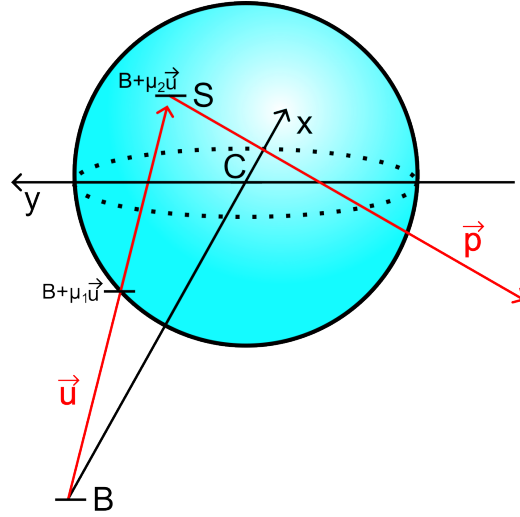


Figure 4: Source of light is outside the sphere

If, on the other hand, the source of light is inside the sphere, as shown in figure 5, the points of intersection are on the opposite sides of the line. The points have coordinates $B + \mu_1 \vec{u}$ and $B + \mu_2 \vec{u}$ and it is clear that either μ_1 or μ_2 must be negative. Since $\mu_1 \leq \mu_2$, we can conclude that μ_1 is negative. Multiplying by a negative scalar is equivalent to flipping the direction of a vector by 180 degrees. In such situation, the light would travel in the opposite direction of its predetermined direction vector. Therefore the point of reflection S is $B + \mu_2 \vec{u}$.

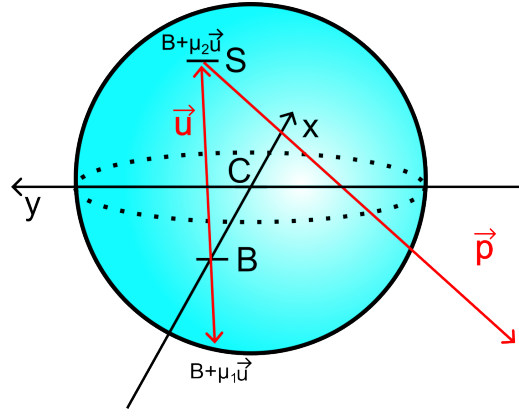


Figure 5: Source of light is inside the sphere

Lastly, if the source of light is on the surface of the sphere as shown in figure 6, coordinates of one of the points of intersection must be identical to the source of light B . Another point of intersection is in the direction of $\vec{u}$, and since $\mu_1 \leq \mu_2$, we can conclude that $\mu_1 = 0$ therefore $B + \mu_1 \vec{u} \equiv B$ and point of reflection S is $B + \mu_2 \vec{u}$.

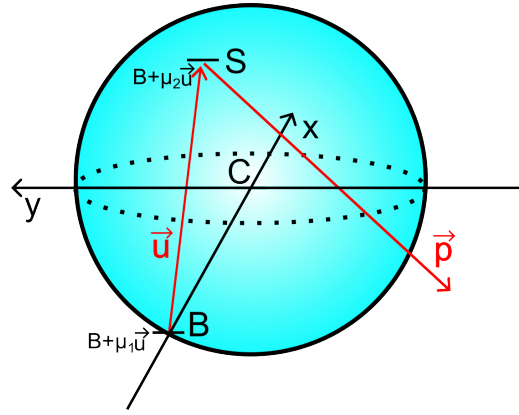


Figure 6: Source of light is on the surface of the sphere

5 Calculating the Direction Vector of Reflected Ray

Our equation for calculating the direction vector of a reflected ray requires knowing the normal vector at the point of reflection. We can utilize the property of a sphere, that normal vector points in the same direction as radius vector, therefore, it points to the center of the sphere, as seen on figure 7. The center is at the origin $O = (0, 0, 0)$. Thus we can calculate the normal vector $\vec{n}$:

$$\vec{n} = \overrightarrow{SO} = O - S = -B - \mu_2 \vec{u}$$

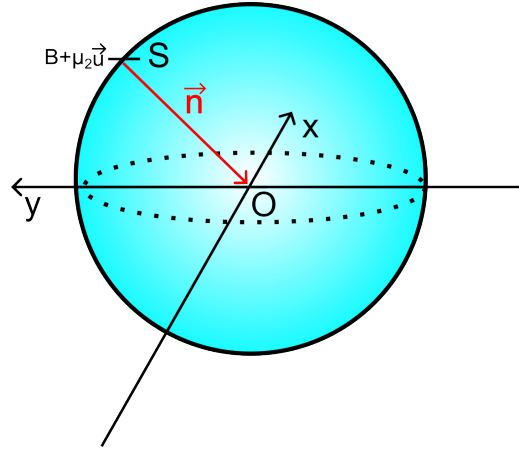


Figure 7: Normal vector at the reflection point

Since this vector represents the distance from the surface of the sphere to the center, its magnitude is 1, therefore we can use it in the equation for calculating direction vector of a reflected ray:

$$\vec{p} = \vec{u} - 2(\vec{u} \cdot \vec{n}) \vec{n}$$

$$\vec{p} = \vec{u} + 2(\vec{u} \cdot (-B - \mu_2 \vec{u})) (B + \mu_2 \vec{u})$$

Firstly, the dot product is evaluated:

$$\vec{u} \cdot (-B - \mu_2 \vec{u}) = \begin{pmatrix} \cos\alpha \\ \cos\beta \\ \cos\gamma \end{pmatrix} \cdot \begin{pmatrix} -B_1 - \mu_2 \cos\alpha \\ -B_2 - \mu_2 \cos\beta \\ -B_3 - \mu_2 \cos\gamma \end{pmatrix}$$

$$\vec{u} \cdot (-B - \mu_2 \vec{u}) = -B_1 \cos\alpha - \mu_2 \cos^2\alpha - B_2 \cos\beta - \mu_2 \cos^2\beta - B_3 \cos\gamma - \mu_2 \cos^2\gamma$$

$$\vec{u} \cdot (-B - \mu_2 \vec{u}) = -B_1 \cos\alpha - B_2 \cos\beta - B_3 \cos\gamma - \mu_2$$

for spatial constraints of the exploration, $\vec{u} \cdot (-B - \mu_2 \vec{u})$ will be denoted as Θ :

$$\vec{p} = \vec{u} + 2\Theta(B + \mu_2 \vec{u})$$

$$\vec{p} = \vec{u} + 2\Theta B + 2\Theta \mu_2 \vec{u}$$

$$\vec{p} = (2\Theta \mu_2 + 1) \vec{u} + 2\Theta B$$

Now we have all the information to complete the $\text{鏡}(B, \vec{u})$ formula for our spherical mirror:

$$\text{鏡}(B, \vec{u}) : (S, \vec{p}) = (B + \mu_2 \vec{u}, (2\Theta\mu_2 + 1)\vec{u} + 2\Theta B)$$

6 Repeated Reflections

Although, we have the explicit formula for the point of reflection and its direction vector, some difficulties can occur, when a light ray is reflected more than once from the sphere, as shown in figure 8. This section discusses how to express these repeated reflections. However, we need to set some constraints beforehand.

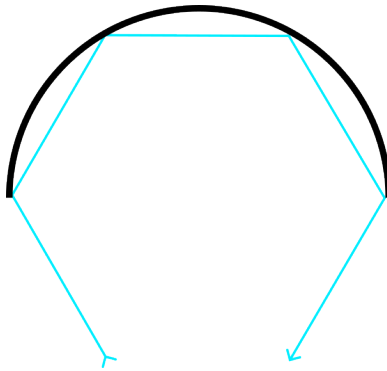


Figure 8: Multiple reflections inside the sphere

It is assumed that the source of light B lies on the X -axis, in other words B has coordinates $(B_1, 0, 0)$. This restriction was made for several reasons. Firstly, if we imagine looking into the spherical mirror, it makes sense to put our line of sight on the X -axis to ensure a complete view of the concave surface of the mirror. Furthermore, by considering all points, we would need to deal with situation, that the view of the reflective surface is blocked - we only see the outside of the mirror. Lastly, it allows for simpler calculations that can be described within the range of this exploration.

The value of B_1 is going to be limited to $B_1 < 0$, again, for practical reasons. Some complications with the calculation of the number of reflections arise, if the source of light is placed "inside" the mirror, therefore, it is assumed the source of light B is outside of the mirror, i.e., $B_1 < 0$.

From the law of reflection, that the initial light ray $\vec{u}$, normal vector $\vec{n}$, and the reflected

light ray $\vec{p}$ lie on the same plane. We can inductively conclude that all the other repeated rays will lie on the same plane as well. Therefore, in visualizations, we will restrict our view on a plane Π defined by the X -axis and the first point of reflection S , as this plane contains the initial light ray vector, normal vector and all the repeated reflected vectors, as shown in figure 9.

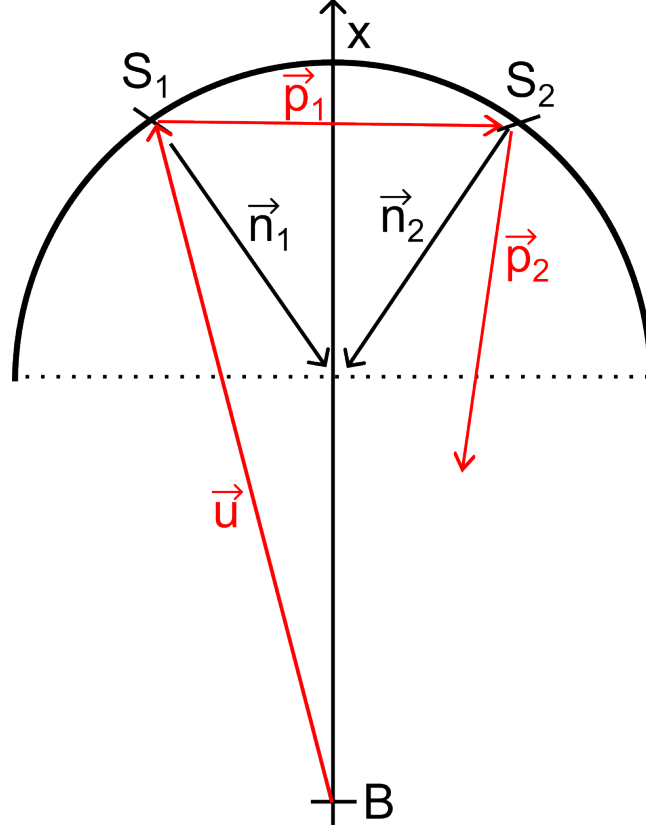


Figure 9: Plane Π defined by X -axis and S , with multiple reflected light rays

We will distinguish individual points of reflection, reflected vectors and normal vectors by using indices, where $S_k, \vec{p}_k, \vec{n}_k$ denote the k -th point of reflection, k -th reflected vector, and k -th normal vector respectively, as seen on figure 9.

In the last section, the point of reflection S and the reflected ray vector $\vec{p}$ were expressed as a function of the initial light ray vector $\vec{u}$ and coordinates of the source of light B :

$$\text{鏡}(B, \vec{u}) : (S, \vec{p}) = (B + \mu_2 \vec{u}, (2\Theta\mu_2 + 1)\vec{u} + 2\Theta B)$$

In figure 9, it can be seen that the pair $(S_1, \vec{p}_1)$ is obtained by $\text{鏡}(B, \vec{u})$. Similarly, the pair $(S_2, \vec{p}_2)$ is calculated by $\text{鏡}(S_1, \vec{p}_1)$. We can imagine, that the light ray starts from the

S_1 and the light ray goes in direction $\vec{p}_1$ and since $\text{鏡}(B, \vec{u})$ calculates the point of reflection and the vector of reflected ray for all positions of light source and all initial light ray vectors (if such point of reflection exists), $\text{鏡}(B, \vec{u})$ can be used for calculating repeated reflections.

Since $(S_1, \vec{p}_1) = \text{鏡}(B, \vec{u})$, we can rewrite $\text{鏡}(S_1, \vec{p}_1)$ as $\text{鏡}(\text{鏡}(B, \vec{u}))$. Hence:

$$(S_2, \vec{p}_2) = \text{鏡}(\text{鏡}(B, \vec{u}))$$

We can inductively extend this logic for all pairs of $(S_n, \vec{p}_n)$:

$$(S_n, \vec{p}_n) = \text{鏡}(S_{n-1}, \vec{p}_{n-1})$$

$$(S_n, \vec{p}_n) = \text{鏡}(\text{鏡}(S_{n-2}, \vec{p}_{n-2}))$$

$$(S_n, \vec{p}_n) = \underbrace{\text{鏡}(\text{鏡}(\dots \text{鏡}(B, \vec{u}))\dots)}_{n \text{ times}}$$

We will use an alternative form to indicate the number of nested functions:

$$(S_n, \vec{p}_n) = \underbrace{\text{鏡}(\text{鏡}(\dots \text{鏡}(B, \vec{u}))\dots)}_{n \text{ times}} \leftrightarrow (S_n, \vec{p}_n) = \text{鏡}^n(B, \vec{u})$$

Note, that for each nesting, a new value for μ_2 and Θ needs to be calculated using the inputs of the function. Therefore, the expression for n -th point of reflection and n -th reflected light vector in terms of B and $\vec{u}$ was found.

7 Determining the Number of Reflections

This section discusses how the position of B (its X -coordinate) and the direction of $\vec{u}$ influence the number of reflections. See figure 10:

We have 2 reflections from the spherical mirror at points S_1, S_2 with reflection vectors $\vec{p}_1, \vec{p}_2$ respectively. Note, that a third point of reflection S_3 is marked on the graph. It lies on the sphere that defines the mirror, but does not satisfy $x > 0$.

P is a point lying in Π , such that its X -coordinate is 0. It is easy to see that P lies on the "edge of the mirror". There are two options for P , each one lying on a different edge. We

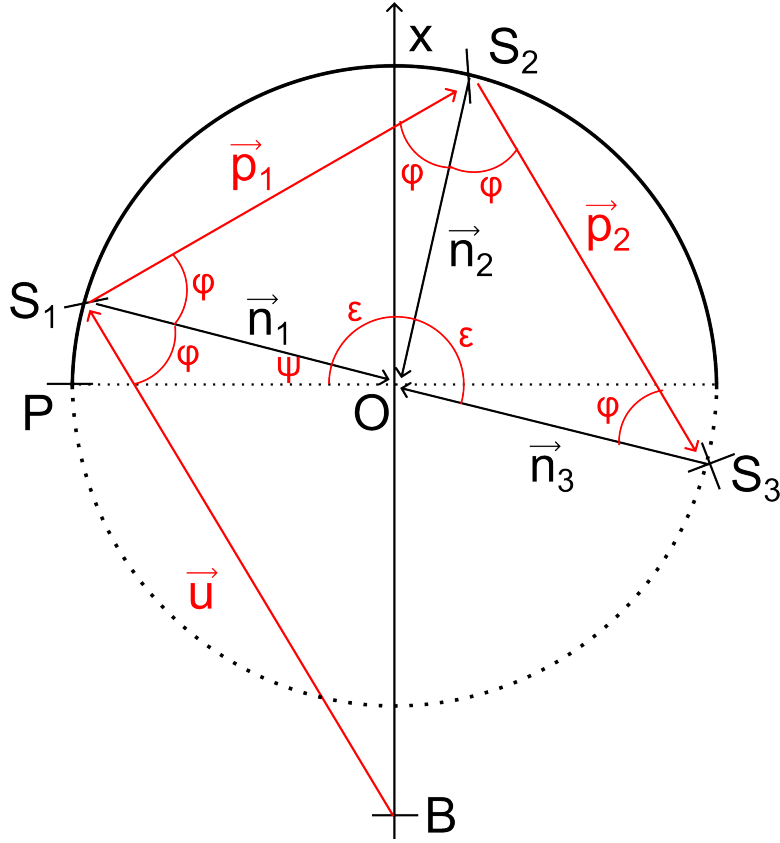


Figure 10: Angles in spherical mirror

define P to be the point closer to S_1 , if both points are equally distant from S_1 , the choice of P is arbitrary. Let $|\angle POS_1| = \psi$

From figure 10, it can be deduced that n -th point of reflection S_n lies on the mirror (will reflect light) if and only if $|\angle POS_n| < \pi$ (we measure the angle that goes "through" the mirror). For example, S_1 will reflect light since $|\angle POS_1| < \pi$, whereas S_3 will not since $|\angle POS_3| > \pi$.

Let φ be $|\angle BS_1O|$. The law of reflection states that also $|\angle OS_1S_2| = \varphi$. $|S_1O| = |S_2O|$, therefore the triangle $\triangle S_1OS_2$ is an isosceles; thus, $|\angle OS_1S_2| = |\angle OS_2S_1| = \varphi$. By similar logic, $|\angle OS_2S_1| = |\angle OS_2S_3| = \varphi$, $|S_2O| = |S_3O|$, thus triangle $\triangle S_2OS_3$ is an isosceles which implies $|\angle OS_2S_3| = |\angle OS_3S_2| = \varphi$. This process can be inductively extended to all points S_n . Therefore, it is true that:

$$\forall n : n \in \mathbb{N}, |\angle OS_n S_{n+1}| = |\angle OS_{n+1} S_n| = |\angle OS_{n+1} S_{n+2}| = |\angle OS_{n+2} S_{n+1}|$$

this implies:

$$|\angle S_n O S_{n+1}| = \pi - |\angle O S_n S_{n+1}| - |\angle O S_{n+1} S_n| = \pi - 2\varphi$$

$\pi - 2\varphi$ will be denoted as ε . We can now return to the inequality:

$$|\angle P O S_n| < \pi$$

Number of repeated reflections from the spherical mirror is the largest whole number n , for which the inequality above is true. $|\angle P O S_n|$ can be rewritten:

$$|\angle P O S_n| = |\angle P O S_1| + |\angle S_1 O S_2| + |\angle S_2 O S_3| \dots + |\angle S_{n-1} O S_n|$$

after substituting ψ for $|\angle P O S_1|$ and ε for each angle in the form $|\angle S_k O S_{k+1}|$:

$$|\angle P O S_n| = \psi + (n - 1)\varepsilon$$

This rewritten form of $|\angle P O S_n|$ can be put into the inequality and we can solve for n :

$$\psi + (n - 1)\varepsilon < \pi$$

$$n - 1 < \frac{\pi - \psi}{\varepsilon}$$

$$n < \frac{\pi - \psi}{\varepsilon} + 1$$

since n is the largest whole number that satisfies the inequality it is equal to the largest whole number not greater than the right side of inequality:

$$n = \left\lfloor \frac{\pi - \psi}{\varepsilon} + 1 \right\rfloor$$

$\lfloor x \rfloor$ is a floor function, which outputs the largest whole number smaller or equal x . Therefore, we have found the expression for the number of reflections n .

8 Calculating the Sizes of Angles

The last two steps of the exploration are expressing angles ψ and ε in terms of variables known to us and combining everything together, which is done in this section.

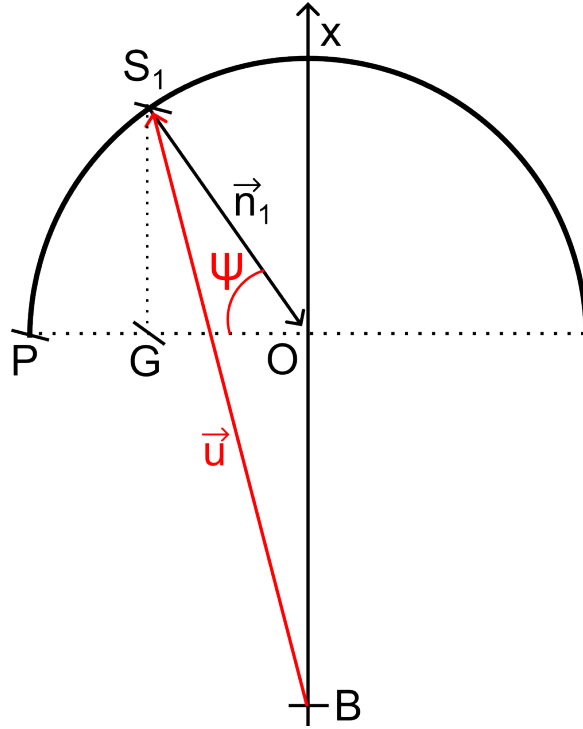


Figure 11: Angle ψ

In figure 11, it can be seen that the distance of the point S_1 and line OP is $\sin\psi$. G is a perpendicular projection of S_1 on the line OP . Line OP is perpendicular to the X -axis and S_1G is perpendicular to the line OP implying S_1G is parallel with the X -axis. Therefore S_1 and G differ only by the X -coordinate and $|S_1G|$ is the difference in their X coordinates.

$$\sin\psi = |S_1G|$$

recalling that the X -coordinates of G and S_1 are 0 and $B_1 + \mu_2 \cos\alpha$ respectively:

$$\sin\psi = B_1 + \mu_2 \cos\alpha - 0$$

$$\sin\psi = B_1 + \mu_2 \cos\alpha$$

$$\psi = \arcsin(B_1 + \mu_2 \cos\alpha)$$

Thus, we have expressed ψ in terms of components of the light source B and initial light ray direction vector $\vec{u}$.

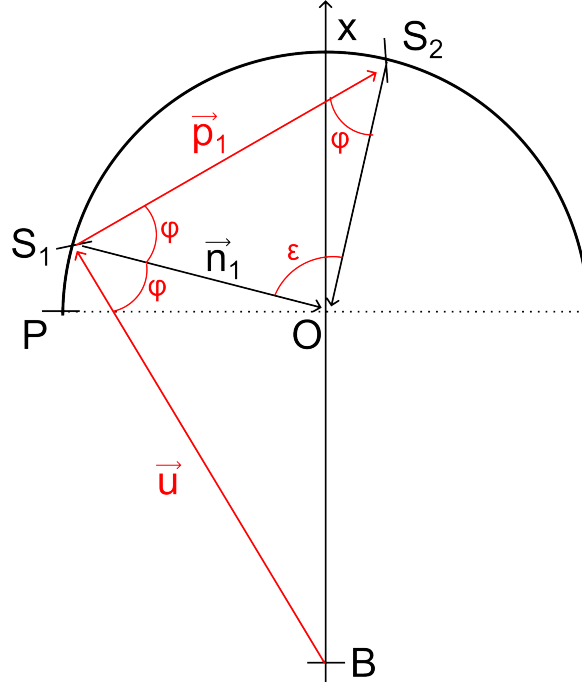


Figure 12: Angle ε

To calculate ε recall that $\varepsilon = \pi - 2\varphi$. This equality can be seen on the figure 12, as the sum of angles in a triangle is π . Since φ is the size of the angle between the normal vector $\vec{n}_1$ and reversed direction vector of the initial light ray $-\vec{u}$:

$$\cos\varphi = -\vec{u} \cdot \vec{n}$$

recall that $\vec{u} \cdot \vec{n} = \vec{u} \cdot (-B - \mu_2 \vec{u}) = \Theta$:

$$\cos\varphi = -\vec{u} \cdot \vec{n} = -\Theta$$

$$\varphi = \arccos(-\Theta)$$

Therefore:

$$\varepsilon = \pi - 2\arccos(-\Theta)$$

By calculating the values for ψ and ε , we have an expression for n that is dependent only

on the source of light and the direction vector of the initial light ray.

$$n = \left\lfloor \frac{\pi - \psi}{\varepsilon} + 1 \right\rfloor$$

$$n = \left\lfloor \frac{\pi - \arcsin(B_1 + \mu_2 \cos \alpha)}{\pi - 2 \arccos(-\Theta)} + 1 \right\rfloor$$

As a result, we have completed the formula for the reflection seen on the spherical mirror with equation $x^2 + y^2 + z^2 = 1 \{x > 0\}$. The final point of reflection and the final direction vector of the reflected will be denoted as $S, \vec{p}$ respectively:

$$(S, \vec{p}) = \text{鏡}^n(B, \vec{u})$$

where the light source $B = (B_1, 0, 0)$ such that $B_1 < 0$ and the initial direction vector $\vec{u} = (\cos \alpha, \cos \beta, \cos \gamma)$:

$$\mu_2 = -(B_1 \cos \alpha) + \sqrt{(B_1 \cos \alpha)^2 - B_1^2 + 1}$$

$$\Theta = \vec{u} \cdot (-B - \mu_2 \vec{u})$$

$$\text{鏡}(B, \vec{u}) = (B + \mu_2 \vec{u}, (2\Theta\mu_2 + 1)\vec{u} + 2\Theta B)$$

$$n = \left\lfloor \frac{\pi - \arcsin(B_1 + \mu_2 \cos \alpha)}{\pi - 2 \arccos(-\Theta)} + 1 \right\rfloor$$

9 Visualization of the Reflections of a Spherical Mirror

In this section, we introduce a visual representation developed for the purpose of this exploration, that models the behavior of the light rays reflected from the mirror.

All of the calculations derived in this exploration were used for the modeling of the spherical mirror visualization. In this model, we are able to modify the position of the light source B and the direction vector of the initial light ray. Based on these parameters, the simulation shows reflected light rays up to 3 reflections, as shown on figure. 13.

Furthermore, the simulation allows us to see, which values of n (number of repeated reflections) are possible for a given point B . It is indicated by regions of different colors on

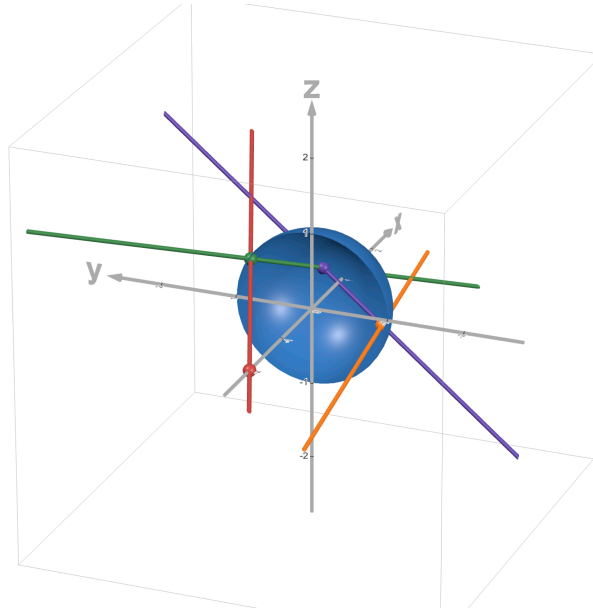


Figure 13: Spherical mirror model with light source B (red dot) and 3 repeated reflections (initial light ray (red), 1st reflection (green), 2nd reflection (purple), 3rd reflection (orange))

the sphere which show the place of the first point of reflection. For example, if the first point of reflection is in the red region, $n = 1$.

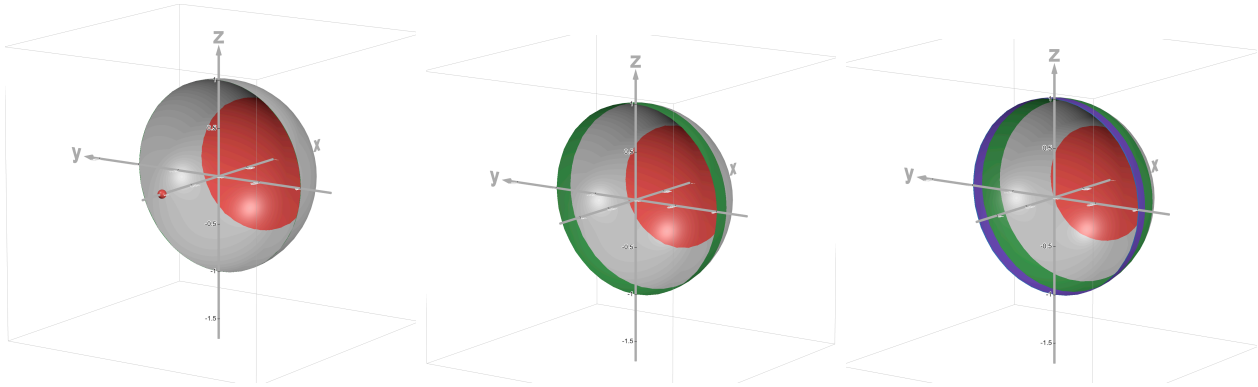


Figure 14: Possible Values of n for $B_1 = -1$, $B_1 = -1.5$, and $B_1 = -2.5$ Respectively

In figure 14, we see that $B_1 = -1$ allows for only 1 reflection (red) or 2 repeated reflections (light gray) to occur. Whereas, $B_1 = -1.5$ 3 reflections (green) are possible, and for $B_1 = -2.5$ even 4 repeated reflections from the mirror can occur. Note, how the number of reflections increases, as we move further from the X -axis.

This model to some extent explains the flipped reflection of my face, I have seen on the surface of spoon. Notice, that a the direction of a light ray reflected from the red area is always flipped, if the initial light ray pointed up and to the right the reflected ray points down and to the left, as shown in figure 15. The proof of this phenomena is outside of scope

of the exploration, but applying this fact for every initial light ray traced from the source B explains the flipped image of my face.

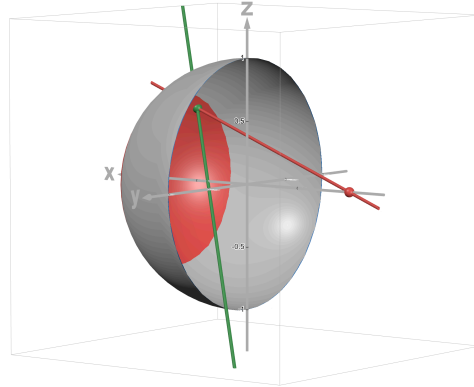


Figure 15: Example of a flipped light ray

The simulation tool is provided in the appendix and I strongly recommend experimenting with various positions of the light source to see their effect on reflections.

10 Conclusion and Evaluation

This exploration aimed to describe the reflections of a spherical mirror by tracing the path of individual light rays and treating them as vectors. A relationship for the reflected light ray expressed in terms of coordinates of the light source and direction vector of the initial light ray was found allowing us to construct virtual models of spherical mirror such as the one seen in the exploration.

However, while the behavior of individual light rays was observed, the exploration did not analyze how do the reflected light rays behave together. As a result, the distortion seen on the concave surface of the spherical mirror could not be explained. Therefore, the way the mirror morphs the image is a subject to further investigation.

While the formula for reflections on the mirror was found, it involves nested functions. Finding a relationship, that calculates the final point of reflection and direction vector of the reflected ray based on the angles created in the spherical mirror should be a subject to further investigation since removing the nested functions reduces the computational complexity of the simulation.

Lastly, due to the restricted scope of the exploration, simplifications were made during the analysis which attributed to limitations of the exploration. The relationship found in the exploration applies only for a limited interval of points lying on the X -axis. As a result, we do not know precisely, how the mirror behaves for light sources not on the X -axis and inside the mirror. Moreover, the relationship applies only for a sphere with radius 1 to simplify the calculations. Therefore, all of the simplifying generalizations should be addressed in the further investigation by including all the emitted variables in the calculations and considering all the cases.

11 References

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12 Appendix

Link to the virtual simulation of spherical mirror: <https://www.desmos.com/3d/vprwpzndif>